

Chapter 8: Protein Language Models (pLMs) & Evolutionary Dynamics

Chapters 1-7 worked almost entirely with small molecules — SMILES strings, molecular graphs, generated ligands. Part III turns to the other half of this book’s title: macromolecules. A protein sequence is, formally, a string over a 20-letter alphabet, and Chapter 3 (Section 3.1) already introduced the classical tools for reading meaning out of that string — substitution matrices, pairwise and multiple sequence alignment. This chapter asks a different question: instead of hand-designing what a sequence’s letters mean, can a model *learn* it, purely from being shown enormous numbers of real protein sequences and asked to predict masked-out pieces of them? The answer — yes, well enough to be scientifically useful with zero task-specific training — is what makes **protein language models** (pLMs) the foundation for every remaining chapter in Part III: Chapter 9’s structure predictors and Chapter 10’s protein designers both build directly on the representations this chapter introduces.

8.1 The Evolutionary Scale Hypothesis

Anfinsen’s (1973) thermodynamic hypothesis — that a protein’s amino acid sequence alone determines its native 3D structure, with no additional biological machinery required — is the foundational premise of computational structural biology: everything Chapters 3, 9, and 10 do rests on it being true. It says the information is *there*, encoded in the sequence. It says nothing about how to decode it.

Evolution has been running that decoding experiment for billions of years. Every sequence in a modern protein family is a descendant of some ancestral sequence, filtered through natural selection: mutations that broke folding or function were eliminated, mutations that were neutral or beneficial persisted. A **multiple sequence alignment** (MSA) of a protein family — Chapter 3, Section 3.1’s tool, revisited here as promised — makes this filtering visible as statistical structure: positions critical to folding or function stay highly conserved across millions of years of divergence, while tolerant positions drift freely; pairs of positions that are physically in contact tend to **co-evolve**, because a destabilizing substitution at one is often compensated by a correlated substitution at the other. Classical unsupervised methods exploit exactly this signal from a *single* family’s MSA: EVcouplings-style direct coupling analysis infers residue-residue contacts from co-evolving pairs, and Riesselman et al.’s (2018) DeepSequence trains a variational autoencoder on one family’s alignment to model that family’s implicit fitness landscape, reporting zero-shot mutation-effect predictions competitive with the individual assays it evaluates against — without ever seeing experimental labels.

The **evolutionary scale hypothesis** — the premise this chapter’s namesake models are built on — is a scale-up of that same idea, with one structural change: instead of an MSA of one family (typically thousands of sequences, aligned to each other’s exact positions), train a single self-supervised model on hundreds of millions of *unaligned* sequences spanning the entire observed protein universe at once. Devlin et al. (2019) introduced the training objective this depends on for language generally — **masked language modeling** (MLM): corrupt a fraction of input tokens by replacing them with a special mask token, and train the model to reconstruct them from context,

$$\mathcal{L}_{\text{MLM}}(\theta) = -\mathbb{E}_{x \sim \mathcal{D}} \sum_{i \in M} \log p_{\theta}(x_i | x_{\setminus M})$$

where M is the randomly chosen set of masked positions in sequence x and $x_{\setminus M}$ denotes the full sequence with those positions replaced by the mask token. Applied to protein sequences with the 20-letter amino acid alphabet in place of a natural-language vocabulary, Rives et al. (2021) showed this objective, trained at sufficient scale (250 million UniParc sequences; up to 650 million parameters), causes structural and functional information to emerge as a *side effect* of the reconstruction task — with no structural label, no alignment, and no family-specific input ever supplied — measurable directly in the trained model’s internal representations (Section 8.3 makes this concrete: masking a real position and reading off the model’s implied preferences over the 20 amino acids at that position is precisely the reconstruction task the model was trained on, applied at inference time as a scoring tool). This is the sense in which “evolutionary scale” replaces “one family’s MSA” as the source of selective-pressure information: not because cross-family evolutionary signal stops mattering, but because a big enough corpus of raw sequences turns out to contain it implicitly, recoverable without ever constructing an alignment.

8.2 pLM Families

ESM-1b (Rives et al., 2021) is a 650M-parameter Transformer *encoder* (bidirectional self-attention, unlike Chapter 7’s causal decoder — every position attends to every other position, masked ones included, which is exactly what reconstructing a masked position from its full surrounding context requires) trained with the MLM objective above on UniRef50. **ESM-2** (Lin et al., 2023) is the direct successor used throughout this chapter’s hands-on project: architecturally the same encoder design, with three concrete changes — rotary positional embeddings in place of ESM-1b’s learned absolute ones, a training-data switch to UniRef50 with UniRef90 sequence sampling for diversity, and, most consequentially, a scale sweep spanning eight model sizes from 8M to 15B parameters, trained on an identical recipe so that model size is isolated as the one varying factor. Lin et al. (2023) report that structure-prediction accuracy (measured via ESMFold, the structure head trained on top of ESM-2’s representations — Chapter 9’s subject) improves smoothly and predictably with scale across this entire sweep, with no sign of saturating even at 15B parameters; Section 8.4 checks whether the same scale-dependence shows up in this chapter’s much smaller zero-shot mutation-scoring task, across the 8M/35M/150M range that fits comfortably in a free Colab CPU runtime.

ProtTrans (Elnaggar et al., 2022) takes a parallel path to the same goal: rather than one architecture, it trains and benchmarks several established NLP language model architectures — BERT, ALBERT, Electra, T5, among others — on protein sequences (UniRef and BFD, up to 393 billion residues), each adapted only by swapping the vocabulary from natural-language tokens to amino acids. ProtBERT, its BERT-architecture member, is the most widely used single output of that effort, sharing ESM-1b’s encoder-plus-MLM design but trained with a different data mixture and codebase. The three families make the same underlying bet — self-supervised reconstruction at scale extracts biologically meaningful structure — with different choices of architecture, training corpus, and (for ESM-2 specifically) an explicit, deliberately isolated study of scale itself; Section

8.4 uses ESM-2 because its published scale sweep is exactly what a scale-comparison hands-on project needs.

8.3 Biological Embeddings

A trained pLM offers two distinct things a downstream task can consume, and this chapter’s hands-on project uses both.

Masked-marginal scoring. Because the MLM objective already trains the model to output $p_\theta(x_i | x_{\setminus\{i\}})$ — a full probability distribution over the 20 amino acids at any position, given everything else — that same forward pass, run once at inference time with no gradient updates at all, can be turned directly into a zero-shot mutation-effect predictor. Meier et al. (2021) formalize this as the **masked-marginal** method: to score a substitution at position i from wild-type residue x_i^{wt} to mutant residue x_i^{mut} , mask position i in the wild-type sequence, run one forward pass, and take the log-odds ratio between the model’s two resulting log-probabilities,

$$\Delta_i(x^{\text{mut}}) = \log p_\theta(x_i^{\text{mut}} | x_{\setminus\{i\}}) - \log p_\theta(x_i^{\text{wt}} | x_{\setminus\{i\}})$$

A positive Δ_i means the model finds the mutant residue *more* plausible than the wild-type one at that position, given everything else in the sequence — evolutionary/structural plausibility, not a direct physical measurement, but one Meier et al. (2021) show correlates with real experimental effects (stability, binding, catalytic activity — the “function” named in this chapter’s title) across dozens of published deep mutational scanning (DMS) datasets, with no per-dataset fine-tuning. Because every mutant at a given position shares the exact same masked forward pass, scoring every possible substitution at a position costs one forward pass, not one per substitution — Section 8.4’s implementation makes this explicit.

Embeddings as features. The same forward pass also exposes each layer’s per-residue hidden-state vectors — the model’s internal, continuous representation of “what this residue means in this context,” pooled (commonly by averaging across the sequence length, Section 8.4’s choice) into one fixed-length vector per protein when a single summary representation is wanted. Unlike masked-marginal scoring, which requires no downstream training at all, embeddings are typically consumed as *features*: frozen or fine-tuned inputs to a separately trained predictor (a small regression head, most commonly) for a task the base pLM was never directly trained on. Section 8.4’s project uses raw embedding distance itself, without a trained head, as a second, independent zero-shot signal — does moving further from the wild-type embedding, in the representation space the model learned, track with a *larger* functional effect, regardless of the effect’s direction?

Both angles generalize past the single case Section 8.4 runs end to end. Meier et al.’s (2021) own benchmark spans binding, stability, and fitness assays across dozens of proteins; Tsuboyama et al. (2023) applied essentially the same masked-marginal principle at much larger scale specifically for **thermal stability** ($\Delta\Delta G$ of folding, not binding), mega-scale-experimentally measuring over 775,000 folding stability values for small domains and their point mutants — the natural larger-scale extension of the “stability” half of this chapter’s title, cited here as a pointer for exactly that reason rather than re-run in Section 8.4, whose scope stays fixed on one complete, tractable mutant population end to end.

Finally, embeddings are not confined to sequence-level tasks. Chapter 9 picks the same ESM-2 representations back up for a different purpose entirely: ESMFold (Lin et al., 2023, introduced in Section 8.2) attaches a structure-prediction head directly on top of them, producing 3D coordinates from a single sequence with no MSA search step at all — the structural payoff of Section 8.1’s “no alignment required” premise, in contrast to AlphaFold2 (Jumper et al., 2021), which achieves its own state-of-the-art accuracy by explicitly building and processing an MSA inside the model (its Evoformer block), the architectural road this chapter’s family of models chose not to take.

8.4 Hands-on Project: Zero-Shot Single-Point Mutation Effect Prediction with ESM-2

The project code lives in this chapter’s folder (`ch08_protein_language_models/`) and puts Section 8.3’s two scoring methods to a direct, real test: given only a pretrained ESM-2 checkpoint and a wild-type sequence — **no GB1-specific training of any kind** — how well do its zero-shot scores track real, experimentally measured mutation effects?

Data: the complete GB1 single-mutant population

GB1, the B1 immunoglobulin-binding domain of Streptococcal protein G, is a 56-residue domain whose binding to the Fc region of IgG is one of the most extensively mutationally characterized protein interactions in the literature. Olson et al. (2014) developed a yeast-display, deep-sequencing assay to measure IgG-Fc binding fitness for variants at four specific positions (39, 40, 41, 54); Wu et al. (2016) applied it to the full combinatorial space of all 20 amino acids at all four positions simultaneously — $20^4 = 160,000$ possible variants — producing, among the double/triple/quadruple mutants their paper’s own epistasis analysis focuses on, the complete population of every possible **single**-point mutant at these four positions: $4 \times 19 = 76$ variants, plus the wild type itself. That completeness matters for this project specifically: `esm_variant_effect.py`’s hands-on result below is not computed from a sample of single mutants, it is computed from literally all of them, so there is no sampling variance between “the 76 mutants used here” and “the single-mutant population.”

`esm_variant_effect.py` downloads this dataset from the FLIP benchmark’s public redistribution (Dallago et al., 2021) of Wu et al.’s (2016) original data, parses each variant’s 4-letter code against the wild-type code (VDGV) to recover the mutated position, wild-type residue, and mutant residue, and extracts the 56-residue GB1 domain sequence itself (the FLIP CSV’s sequence column appends an unrelated ~200-residue yeast-display fusion tag after position 56, stripped here since it plays no role in GB1’s own fold or binding site).

```
def _row_to_mutant(row: dict) -> SingleMutant | None:
    variant_code = row["Variants"]
    if variant_code == GB1_WT_VARIANT:
        return None
    diffs = [i for i in range(4) if variant_code[i] != GB1_WT_VARIANT[i]]
    idx = diffs[0]
    position = GB1_MUTATED_POSITIONS[idx]
    ...
```

Scoring: masked marginals and embedding distance

ESM2VariantScorer wraps a Hugging Face transformers ESM-2 checkpoint and implements both of Section 8.3’s methods. Masked-marginal scoring exploits the position-sharing efficiency named there directly: GB1’s 76 single mutants touch only 4 distinct positions, so `score_single_mutants` masks and scores each of those 4 positions exactly once — not once per mutant — reusing each position’s resulting 20-amino-acid log-probability distribution for every mutant that shares it.

```
def score_single_mutants(self, wt_sequence, mutants):
    positions = sorted({m.position for m in mutants})
    log_probs_by_position = {
        pos: self.masked_marginal_log_probs(wt_sequence, pos) for pos in
positions
    }
    return np.array([
        log_probs_by_position[m.position][m.mut_aa] -
log_probs_by_position[m.position][m.wt_aa]
        for m in mutants
    ])
```

`embed_sequences` runs one forward pass per full mutant sequence (76 of them, plus wild type — mean-pooling excludes the `<cls>/<eos>` special tokens explicitly rather than averaging over them) and `evaluate_embedding_perturbation` compares each mutant’s cosine distance from the wild-type embedding against its experimental fitness deviation from 1.0 (wild-type-normalized fitness).

Running it and reading the results

```
cd ch08_protein_language_models
pip install -r requirements.txt
python esm_variant_effect.py --use-cached-raw
```

Run against three ESM-2 checkpoints spanning roughly a 20x parameter range (8M / 35M / 150M), scoring all 76 single mutants with both methods, Spearman rank correlation against real experimental fitness came out as follows:

Model	Params	Masked- marginal ρ (n=76)	p-value	Embedding- distance ρ	p-value
ESM-2 8M	7.5M	0.095	0.415	0.111	0.339
ESM-2 35M	33.5M	0.184	0.112	0.087	0.453
ESM-2 150M	148.1M	0.208	0.072	0.314	0.0057

Two things are worth reading precisely here, not just the top line. First, **masked-marginal correlation increases monotonically with model scale** (0.095 -> 0.184 -> 0.208) — directionally consistent with Lin et al.’s (2023) scale-dependent accuracy trend (Section 8.2) — but at $n = 76$, none of the three individually clears the conventional $p < 0.05$ significance threshold; the

150M model comes closest ($p = 0.072$). This is reported as what it is: a suggestive trend, not a statistically confirmed one, at this sample size. Second, the **embedding-distance method tells a sharper story at the largest scale tested**: the 150M model’s correlation between representational shift and functional deviation ($\rho = 0.314$, $p = 0.0057$) is the one result in this table that clears significance outright, while the two smaller models do not approach it. Restricting either analysis to the 27-mutant high-read-count subset (`keep=True` in the source data, a sequencing-depth quality flag, not a fitness-reliability one) produces correlations scattered close to zero at every scale — an expected consequence of cutting the sample size by nearly two-thirds, not a contradiction of the full-population result above; it is not treated as a separate finding here.

As a single concrete example: mutant IDGV (position 39, wild-type valine to isoleucine) has real experimental fitness 1.446 — nearly 45% *better* binding than wild type — and the 150M model’s masked-marginal score for V->I at position 39 is positive, correctly ranking it above wild type; not every one of the 76 mutants is ranked this cleanly, which is exactly what a $\rho \approx 0.2$ correlation, rather than $\rho \approx 1.0$, means in practice.

Why is the correlation modest rather than strong? GB1’s assay measures one specific, engineered property — IgG-Fc binding affinity — at four co-varied positions chosen because mutating them tunes that specific interface, not because they are representative of general sequence conservation. A masked-marginal score reflects general evolutionary/ structural plausibility learned from the training corpus at large; it is not specifically trained to predict *this* binding interaction, and Meier et al. (2021) report exactly this kind of heterogeneity across their own multi-assay benchmark — some DMS datasets correlate strongly with zero-shot pLM scores, others, including affinity-tuning assays at a small number of interface positions, considerably less so. The honest lesson Section 8.4 is built to teach is this one: zero-shot pLM scoring is a genuinely useful, free, no-training-required signal, but not a universal oracle — validating it against real data for the specific property at hand, as done here, is the point, not an afterthought.

Reproducibility

Dependencies are version-floored (`torch>=2.2`, `transformers>=4.40`, `scipy>=1.10` in `requirements.txt`, validated against torch 2.13.0 and transformers 5.15.1 on Python 3.12). `data/gb1_single_mutants_sample.csv` bundles the real, complete wild-type + 76-single-mutant population (extracted from the live FLIP download on 2026-08-20), so the pipeline runs offline and deterministically with `--use-cached-raw` — verified directly: `--use-cached-raw` and a fresh live download of the full 149,361-row dataset produce bit-identical Spearman correlations, since both extract the same 77 rows. All three ESM-2 checkpoints total a few hundred MB on first download from the Hugging Face Hub; with them cached, the full run (3 models x 2 methods x 76 mutants) takes about a minute on a single CPU core, no GPU required. The 16-test suite in `tests/test_esm_variant_effect.py` checks data parsing against the real bundled fixture (including one spot-checked published fitness value), the download function against a mocked in-memory zip, and both scoring methods’ mechanics — including the position-sharing efficiency claim above, checked directly by counting calls — against a tiny, untrained ESM architecture built from the real ESM-2 tokenizer vocabulary, so the suite never downloads a pretrained checkpoint and completes

in under a minute, fully offline. `pip install -r requirements-dev.txt && pytest` reproduces all 16 results.

Limitations and what comes next

This project deliberately stays inside the single-family, single-assay case: one 56-residue domain, one binding readout, exhaustively enumerated at only 4 positions. Extending it to genome-scale variant effect prediction (Section 8.3’s forward reference to Tsuboyama et al., 2023) or to positions outside this specific interface would need a different DMS dataset entirely, not a code change — the pipeline’s data-loading layer is specific to this CSV’s schema, again a deliberate scope decision named here rather than left implicit. The masked-marginal method also stays firmly zero-shot throughout: nothing in this chapter fits a regression head on top of ESM-2 embeddings against GB1 labels, which — per Section 8.3 — is the more common way embeddings get used in practice and would very likely outperform the zero-shot scores reported above, at the cost of needing labeled training data for the specific property being predicted, which is exactly the dependency zero-shot scoring is valuable for not requiring. Finally, everything in this chapter operates on sequence alone; Chapter 9 picks these same ESM-2 representations back up to predict the 3D structures those sequences fold into.

A note on Google Colab

Colab’s default runtime preinstalls `torch` but not `transformers` or `scipy`; run `!pip install transformers scipy` in the first cell. No GPU is required for the model sizes used here (8M-150M parameters); larger ESM-2 checkpoints (650M and up) would benefit from one.

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See Chapter 7’s references for Vaswani et al. (2017, the Transformer architecture underlying every pLM in this chapter). RDKit, DeepChem, and the ChEMBL/PDB APIs used in Chapters 1-7 play no role in this chapter; ESM-2’s model weights and tokenizer are distributed through the Hugging Face Hub (facebook/esm2_t6_8M_UR50D, facebook/esm2_t12_35M_UR50D, facebook/esm2_t30_150M_UR50D), and the GB1 dataset is redistributed by the FLIP benchmark under an AFL-3 license (original data CC BY 4.0 via eLife), both confirmed directly from their respective repositories rather than assumed.

All correlations, p-values, and the worked IDGV example cited in Section 8.4 were computed directly by running `esm_variant_effect.py` against the live-downloaded GB1 dataset on 2026-08-20, not taken from a secondary source, and independently reproduced against the bundled offline fixture — see `data/gb1_single_mutants_sample.csv` and `results/esm2_gb1_results.json` to reproduce.